Algorithmic Bias Audit Report

Target File: `./test/approve_sensitive_project_access_.py`

Audit Model: FATE-Auditor-v1

Analysis Type: Static Code FATE Analysis

Status: 4 Flags Detected

Executive Summary

This report details instances of algorithmic bias identified within the source code during automated static analysis. The evaluation surfaced exclusionary logic based on protected attributes and/or proxies, which may propagate differential treatment through the Software Development Life Cycle (SDLC).

Detailed Findings

Finding 1: Proxy Bias

Association: POSITIVE

Non-Protected Attribute: public_reputation_risk

Biased Code Snippet:

```
employee["public_reputation_risk"] == "low"
```

Issue Description: The function uses 'public_reputation_risk' as a proxy for sensitive attributes like socioeconomic status or race, as these can be unfairly correlated with how risk is assessed, leading to preferential outcomes for certain groups.

Impact Reasoning: Relies on a measurable attribute ('public_reputation_risk') as a stand-in for a trait that correlates with a protected group, unfairly impacting groups where that attribute does not accurately indicate the intended outcome.

Finding 2: Demographic Bias

Association: POSITIVE

Non-Protected Attribute: public_reputation_risk

Biased Code Snippet:

```
if employee["public_reputation_risk"] == "low":
```

Issue Description: The function provides a favorable outcome (project approval) only to a specific group defined by the non-protected attribute 'public_reputation_risk', resulting in differential treatment.

Impact Reasoning: Uses a non-protected attribute ('public_reputation_risk') to result in differential treatment of individuals based on group membership or proxy grouping.

Finding 3: Threshold/Cutoff Bias

Association: POSITIVE

Non-Protected Attribute: public_reputation_risk

Biased Code Snippet:

```
employee["public_reputation_risk"] == "low"
```

Issue Description: The function uses a rigid categorical cutoff ('low') on the 'public_reputation_risk' attribute to determine eligibility for project access, creating a sharp boundary that may unfairly exclude individuals.

Impact Reasoning: Uses a fixed non-protected attribute ('public_reputation_risk == "low"') with a threshold to determine eligibility or outcomes, where the rigid cutoff may produce biased treatment for individuals near the boundary.

Finding 4: Representational Bias

Association: POSITIVE

Non-Protected Attribute: public_reputation_risk

Biased Code Snippet:

```
if employee["public_reputation_risk"] == "low":
            approved.append(employee)
```

Issue Description: The code creates an 'approved' list that exclusively includes individuals with 'low' public reputation risk, thus not representing anyone outside this group in a context of positive career opportunity.

Impact Reasoning: Does not represent those outside of the non-protected attribute (‘public_reputation_risk == "low"') in a context where all should have equal opportunity for inclusion.